

PRINCE2 | P2-22

End Stage Report

Performance of the stage and the case for continuing

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	End Stage Report
Document reference	P2-22-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project Manager
Approver	Project Board (Authorise a Stage or Exception Plan)
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
P2-11	Plan	Next Stage Plan accompanies this report
P2-04	Business Case	Updated position
P2-23	End Project Report	Final-stage equivalent

How to use this template

Aspect	Guidance
Purpose	Provides the Project Board with a summary of stage performance, the current project status, and the forecast for the next stage, so it can decide whether to authorise continuation.
Framework	PRINCE2 — Produced in Managing a Stage Boundary
Created when	At the end of each management stage
Reviewed / updated	At the end stage assessment
Owner	Project Manager
Approver	Project Board (Authorise a Stage or Exception Plan)
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	Scale the depth of each section to the scale, risk and complexity of the work. Delete sections that add no value and record the tailoring decision in the governance approach.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- It reports actual performance against the stage baseline, with explanations for variance
- The updated Business Case position is stated and the continued justification is explicit
- It gives the Board what it needs to make a go/no-go decision, not a narrative of activity
- Follow-on action recommendations are captured for anything unresolved

Contents

Document control.....	2
Version history.....	2
Reviewers.....	2
Approval.....	2
Related products.....	2
How to use this template.....	3
Quality criteria.....	3
1. Report identification.....	5
2. Project Manager's report.....	5
3. Review of the Business Case.....	5
4. Review of project objectives.....	6
5. Review of stage objectives and performance.....	6
6. Review of team performance.....	8
7. Review of products.....	8
Products delivered in this stage.....	8
Products approved with concessions.....	9
Products not delivered.....	9
Off-specifications.....	9
8. Risk summary.....	10
9. Issue and change summary.....	10
10. Lessons report.....	11
11. Forecast for the next stage.....	11
12. Follow-on action recommendations.....	11
13. Board decision.....	12

1. Report identification

Field	Detail
Project	<i>WS1 Facilities and power</i>
Stage reported	<i>D. Okonkwo, Programme Manager</i>
Stage start / end date	<i>Helios AI Platform ART</i>
Report date	<i>18 Sep 2026</i>
Prepared by	<i>D. Okonkwo, Programme Manager</i>
Next stage	<i>Tranche 2 — Scale build</i>

2. Project Manager's report

A short narrative summary of stage performance, including anything the Board should understand that the tables do not convey.

Programme Helios — Tranche 2 scale build, DXB-1 halls A and B. Constrained by the 6.2 MVA site limit at DXB-1 until 2028. Detail is maintained by the Project Manager and reviewed at the monthly Programme Board.

3. Review of the Business Case

Aspect	At stage start	At stage end	Commentary
Total forecast cost	<i>AED 118,000,000 for Tranche 2; tolerance of +5% before escalation</i>	<i>Tranche 2 — Scale build</i>	<i>AED 118,000,000 for Tranche 2; tolerance of +5% before escalation</i>
Total forecast benefit	<i>No more than 10% variance against the benefit profile before escalation</i>	<i>Tranche 2 — Scale build</i>	<i>No more than 10% variance against the benefit profile before escalation</i>
Net benefit	<i>No more than 10% variance against the benefit profile before escalation</i>	<i>Tranche 2 — Scale build</i>	<i>No more than 10% variance against the benefit profile before escalation</i>
Payback / return	<i>Helios AI Platform ART</i>	<i>Tranche 2 — Scale build</i>	<i>Improved since last period following the CDU set-point retune</i>
Aggregate risk exposure	<i>No individual risk may remain above a residual score of 12 without SRO acceptance</i>	<i>Tranche 2 — Scale build</i>	<i>No individual risk may remain above a residual score of 12 without SRO acceptance</i>
Still desirable?	<i>Helios AI Platform ART</i>	<i>Tranche 2 — Scale build</i>	<i>Dependent on the Enterprise Identity Programme; outside programme control</i>
Still viable?	<i>Helios AI Platform</i>	<i>Tranche 2 — Scale</i>	<i>Escalated to the Programme Board on</i>

Aspect	At stage start	At stage end	Commentary
	ART	build	18 Sep 2026; decision requested
Still achievable?	Helios AI Platform ART	Tranche 2 — Scale build	On plan; no intervention required this period

4. Review of project objectives

Objective	Target	Position at stage end	Assessment
Three tenants can run concurrently on hall A with fair-share quota enforced under contention	Zero open critical findings	23 Apr 2027	Delivered and demonstrated at the PI 4 system demo
Low-priority training jobs can be pre-empted and resumed without loss of work	92% at 256-GPU job scale	28 May 2027	Partially met — resume works for framework-native checkpointing only
Only signed container images from the internal registry can run on the estate	1.21 PUE	25 Jun 2027	Delivered; closes a Tranche 1 assurance finding
Nodes failing firmware attestation are quarantined automatically at admission	72% by Q4 2027	18 Sep 2026	Not started — deferred to PI 5 following the vendor attestation API change
Each tenant receives a monthly consumption statement that reconciles to telemetry	3 days median	15 Oct 2026	Delivered; first statements issued to Finance in November
GPU telemetry is available at 10-second resolution with 13-month retention	AED 5.1m per year	30 Nov 2026	Delivered; underpins the utilisation benefit measure

5. Review of stage objectives and performance

Dimension	Stage baseline	Actual	Variance	Within tolerance?	Explanation
Time	Tranche 2 — Scale build	90 days	90 days	3 working days	10 working days
Cost	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation
Scope	Defined by the	In scope:	In scope:	In scope: the	In scope: the physical

Dimension	Stage baseline	Actual	Variance	Within tolerance?	Explanation
	Tranche 2 scope statement; changed only by approved change request	the physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.	the physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.	physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.	infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.
Quality	No open critical or high security findings; acceptance benchmarks met in full	No open critical or high security findings; acceptance benchmarks	No open critical or high security findings; acceptance	No open critical or high security findings; acceptance benchmarks met in full	No open critical or high security findings; acceptance benchmarks met in full

Dimension	Stage baseline	Actual	Variance	Within tolerance?	Explanation
		met in full	benchmarks met in full		
Risk	No individual risk may remain above a residual score of 12 without SRO acceptance	No individual risk may remain above a residual score of 12 without SRO acceptance	No individual risk may remain above a residual score of 12 without SRO acceptance	No individual risk may remain above a residual score of 12 without SRO acceptance	No individual risk may remain above a residual score of 12 without SRO acceptance
Benefit	No more than 10% variance against the benefit profile before escalation	No more than 10% variance against the benefit profile before escalation	No more than 10% variance against the benefit profile before escalation	No more than 10% variance against the benefit profile before escalation	No more than 10% variance against the benefit profile before escalation

6. Review of team performance

Performance of internal teams and suppliers. Factual and constructive.

Performance is measured against four headline indicators: cost and schedule variance against the Tranche 2 baseline, GPU utilisation as a rolling 30-day average, all-reduce efficiency at 256-GPU job scale, and the number of tenants onboarded against the migration commitment. All four are reported monthly to the Programme Board from the observability platform rather than from manual returns.

7. Review of products

Products delivered in this stage

Product ID	Product	Approved by	Date	Quality result
Platform v1.0 — scheduling, quota, observability and image supply chain	Hall A accepted at 1,024 GPUs with acceptance certificate	D. Okonkwo, Release Train Engineer	Mar 2026 – Aug 2026	Closed with no residual exposure
400G fabric commissioning	Parallel filesystem accepted at 1.2 TB/s sustained	L. Haddad, CISO	Sep 2026 – Jun 2027	Escalated to the Programme Board and resolved on 18 Sep

Product ID	Product	Approved by	Date	Quality result
ned and NCCL-validated				2026
Independent security assurance report with findings closed	Operations runbook set with two completed live drills	E. Santos, Procurement Lead	Jul 2027 – Jun 2028	Validated — no change required to the baseline

Products approved with concessions

Product	Concession	Approved by	Follow-up action
Platform v1.0 — scheduling, quota, observability and image supply chain	Operations runbook set with two completed live drills	T. Nakamura, Chief Platform Architect	Enforce mutual TLS on the cluster API and retest by 22 Sep 2026
Hall A accepted at 1,024 GPUs with acceptance certificate	Parallel filesystem accepted at 1.2 TB/s sustained	F. Mansour, Business Change Lead	Broker all vendor access through the monitored in-country jump host
Independent security assurance report with findings closed	400G fabric commissioned and NCCL-validated	D. Okonkwo, Programme Manager	Secure written migration commitments from three business units

Products not delivered

Product	Reason	Revised plan
Parallel filesystem accepted at 1.2 TB/s sustained	Grid capacity at DXB-1 supports Tranche 2 in full, so a second site is not yet justified	Parallel filesystem accepted at 1.2 TB/s sustained
Hall A accepted at 1,024 GPUs with acceptance certificate	Physical lead times do not tolerate emerging requirements, so this element is frozen before order	400G fabric commissioned and NCCL-validated
Platform v1.0 — scheduling, quota, observability and image supply chain	Two independent schedulers fragment the GPU pool and reduce effective capacity	Parallel filesystem accepted at 1.2 TB/s sustained

Off-specifications

Product	Off-specification	Issue ref	Decision
Platform v1.0 — scheduling, quota, observability and image supply chain	Agreed with the integration partner at design freeze, 15 Oct 2026	Excluded from Tranche 2 and deferred to the AUH-2 second site	Approved

Product	Off-specification	Issue ref	Decision
400G fabric commissioned and NCCL-validated	Owned within WS4 Platform and orchestration	Reviewed at each PI boundary by the Release Train Engineer	Rejected
Parallel filesystem accepted at 1.2 TB/s sustained	Confirmed with the FM provider under the existing facilities contract	Escalated to the Portfolio Board where it exceeds programme tolerance	Deferred to the next Board

8. Risk summary

Measure	Stage start	Stage end	Commentary
Open risks	No individual risk may remain above a residual score of 12 without SRO acceptance	Tranche 2 — Scale build	No individual risk may remain above a residual score of 12 without SRO acceptance
New risks this stage	No individual risk may remain above a residual score of 12 without SRO acceptance	Tranche 2 — Scale build	Tranche 2 — Scale build
Risks closed	No individual risk may remain above a residual score of 12 without SRO acceptance	Tranche 2 — Scale build	No individual risk may remain above a residual score of 12 without SRO acceptance
Aggregate exposure	Helios AI Platform ART	Tranche 2 — Scale build	Weak tenant isolation exposes model weights across business units (RSK-003)
Risk budget drawn	No individual risk may remain above a residual score of 12 without SRO acceptance	Tranche 2 — Scale build	No individual risk may remain above a residual score of 12 without SRO acceptance

9. Issue and change summary

Measure	Count	Commentary
Issues raised this stage	Tranche 2 — Scale build	Tranche 2 — Scale build
Issues closed	3	26 Feb 2027
Issues open at stage end	Tranche 2 — Scale build	Tranche 2 — Scale build
Changes approved	8	Integration partner claim for additional structural works
Change budget drawn	AED 17,800,000	AED 240,000

10. Lessons report

Include or reference the Lessons Report (P2-26) if one is being issued at this boundary.

Three themes have emerged from Tranche 1: security assurance run only at completion finds expensive problems late; specialist vendor performance engineering is worth budgeting as a planned activity rather than a remedial one; and stakeholder attitude assessed from a single early conversation is unreliable where autonomy is being removed.

11. Forecast for the next stage

Summarise the next Stage Plan: objectives, products, cost, duration, key risks, and the tolerances being requested.

Programme Helios — Tranche 2 scale build, DXB-1 halls A and B. Dependent on the Enterprise Identity Programme (DEP-003). Detail is maintained by the Project Manager and reviewed at the monthly Programme Board.

12. Follow-on action recommendations

Ref	Recommendation	Reason	Recommended owner
FOA-01	15 Dec 2026	Air cooling would cap density at 12 kW per rack and require 2.4 times the floor area	T. Nakamura, Chief Platform Architect
FOA-02	31 Jan 2027	Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	F. Mansour, Business Change Lead
FOA-03	26 Feb 2027	Supply chain provenance for AI tooling is weak; admission control is the only enforceable point	D. Okonkwo, Programme Manager
FOA-04	31 Mar 2027	Grid capacity at DXB-1 supports Tranche 2 in full, so a second site is not yet justified	K. Adeyemi, Security Architect

13. Board decision

Decision	Detail
Stage approved as complete?	<i>Tranche 2 — Scale build</i>
Next Stage Plan approved?	<i>Tranche 2 — Scale build</i>
Tolerances set for next stage	<i>Tranche 2 — Scale build</i>
Conditions attached	<i>Performance bond of 10%, step-in rights, and design documentation held in escrow</i>
Date of decision	<i>18 Sep 2026</i>